

Operation Manual(EN)

2025.04_A2

OMRON Corporation
Sensor Division, Product Business Div HQ
Industrial Automation Business Company

Introduction

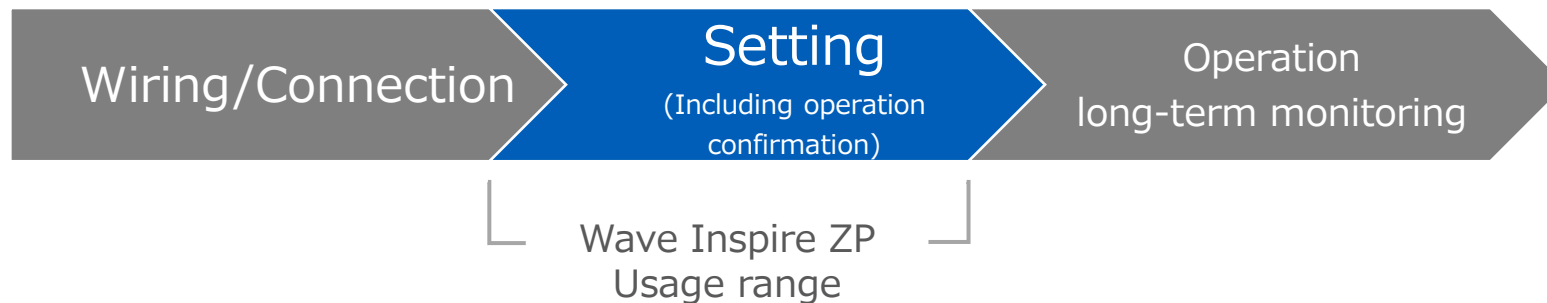
- 1. System Requirements**
- 2. Environment Configuration and Connection Method**
- 3. Startup Screen / Project Creation**
- 4. Overview of Functions and Screen Operations**
- 5. Cautions and Troubleshooting**

Introduction

Wave Inspire ZP (“the tool”) makes monitoring the status easy. In addition to configuring the sensors, it allows you to graph and log the measurement values and judgment results and to display a list of sensor operating statuses.

- Scope of operation of the tool

The tool has a logging function that collects and saves measurement data. However, this function is intended for configuration support and short-term monitoring and it is not intended for long-term operation. If you want to collect data over a long period of time, create and use a system that suits your needs.



※ Microsoft product screen shots reprinted with permission from Microsoft.

1. System Requirements

System Requirements

The system requirements for the PC tool are shown in the table below.

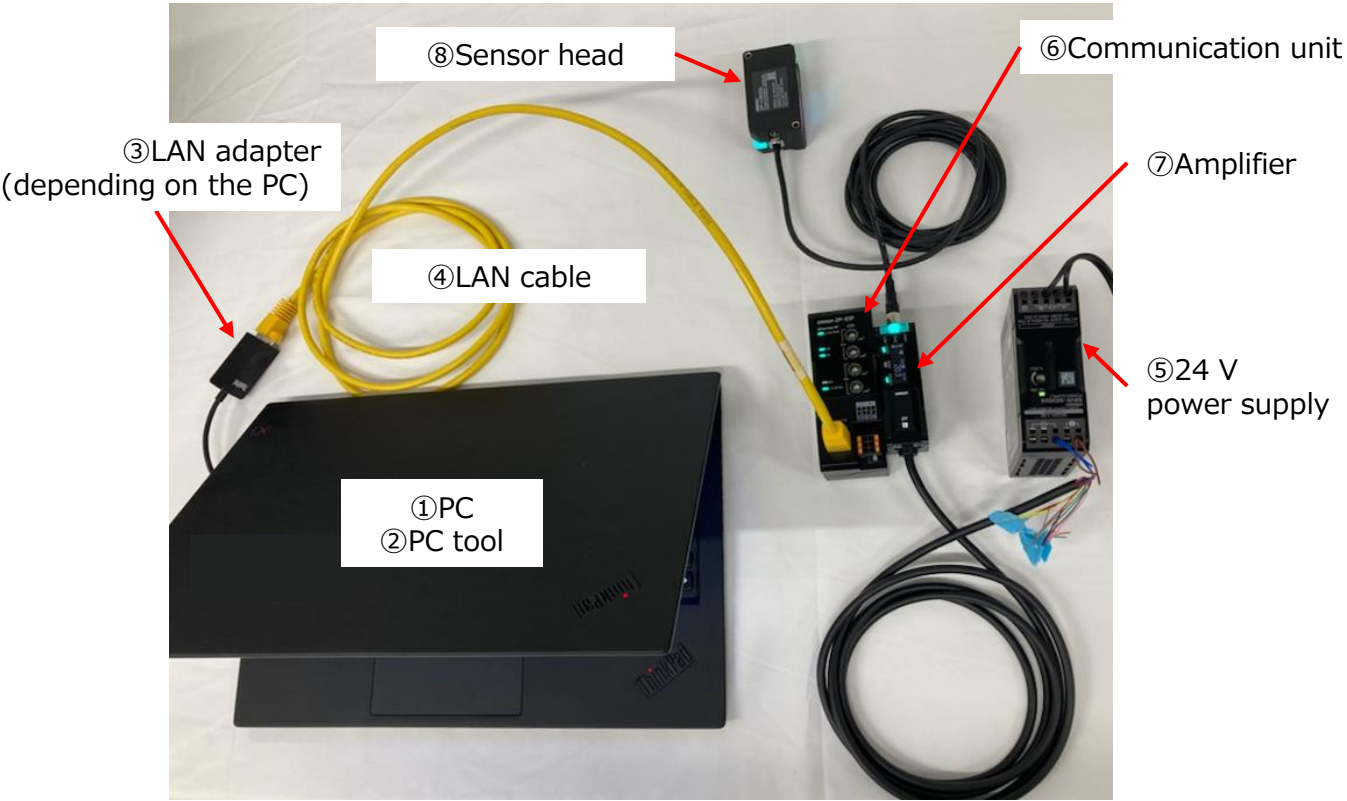
No.	Item	Condition
1	OS	Windows10 64bit / Windows11 64bit
2	CPU	Intel® Core(TM) i5-6200U CPU 2.30GHz min.
3	Disk capacity	10 GB min.
4	Display size	1920 x 1080 (Full HD) min.
5	Network interface	Ethernet port

2. Environment Configuration and Connection Method

Items Required and Connection Configuration

Connect using the configuration below.

	Name	Example Format
①	PC	products on the market
②	PC tool Wave Inspire ZP (Free)	—
③	LAN adapter	products on the market
④	LAN cable	products on the market
⑤	Power supply	S8VK-S06024 etc.
⑥	Communication unit	ZP-EIP
⑦	Amplifier	ZP-L3000 etc.
⑧	Sensor head	ZP-LS050S 2M etc.

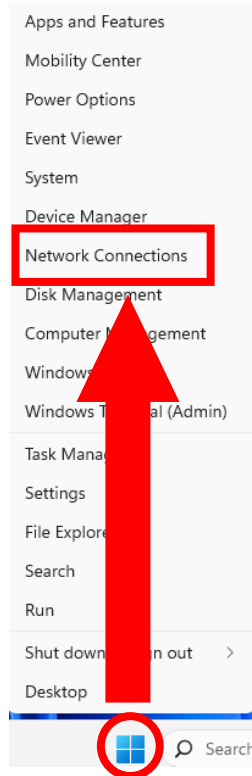


2. Connection Method – Network Adapter Setup (1/4) –

The following operations should be performed by a user with administrator privileges.

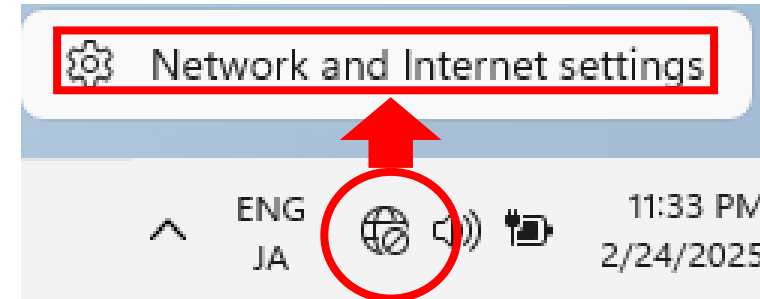
1. Use one of the methods below to open the “Network status” screen.

Right-click the Start menu (square button at the bottom left) and click [Network Connections].



Right-click

Right-click the Network Status icon shown below on the taskbar indicator and click [Open Network & Internet settings].

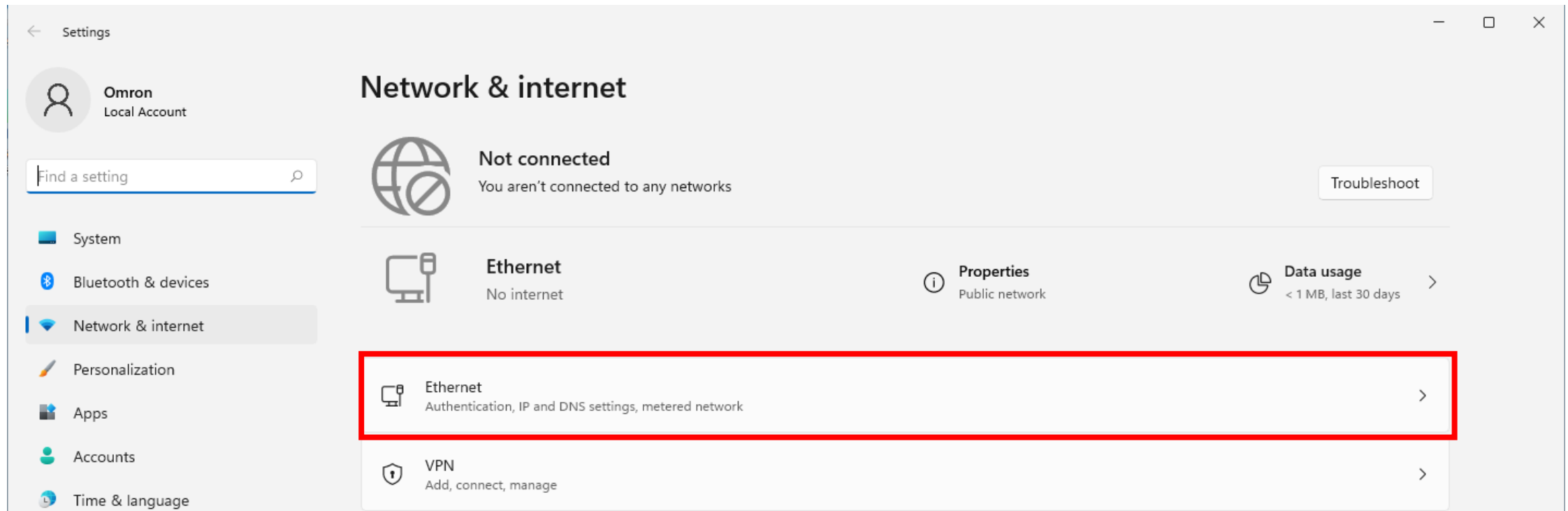


Right-click

2. Connection Method – Network Adapter Setup (2/4)-

The following operations should be performed by a user with administrator privileges.

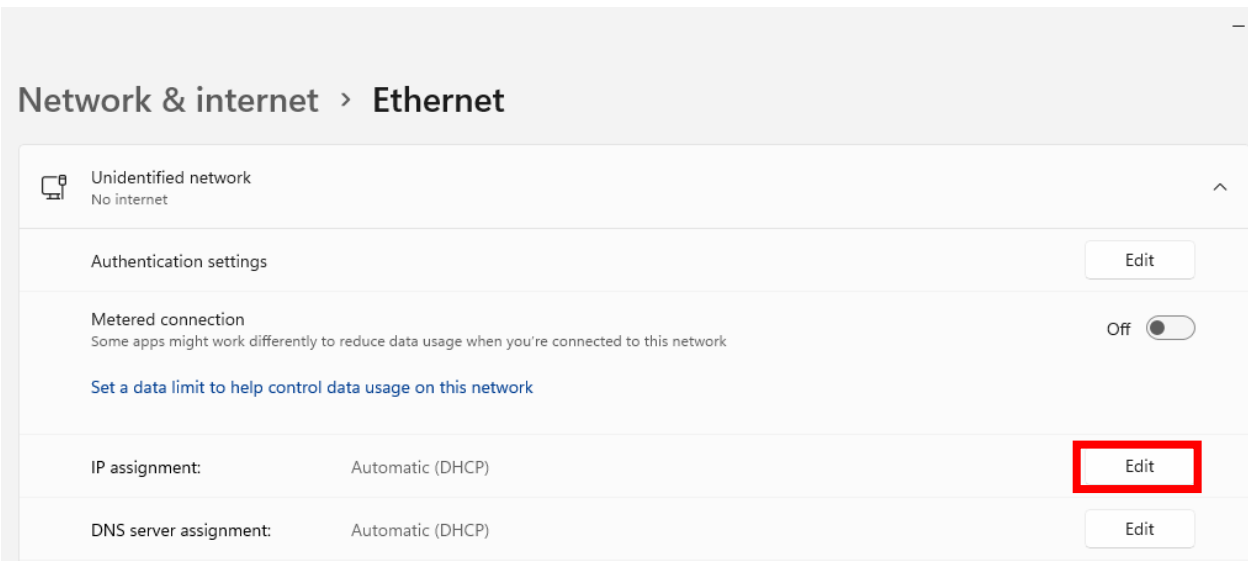
2. When the following screen appears, click [Ethernet].



2. Connection Method – Network Adapter Setup (3/4)-

The following operations should be performed by a user with administrator privileges.

3. Click [Ethernet *] that appears when you insert the wired LAN adapter, and click [Edit] under [IP Assignment].



- * Depending on the device, the number may be different, such as "Ethernet 2"
- * If you don't know which one to select, unplug the wired LAN adapter, wait 2-3 seconds, plug it back in, and select "Ethernet + Number" that appears.

2. Connection Method – Network Adapter Setup (4/4)-

The following operations should be performed by a user with administrator privileges.

4. Set the IP address on the PC side.

The initial IP address of the communication unit is "192.168.250.1".
As an example, "192.168.250.200" is set.

Change [Automatic (DHCP)] to [Manual] in
[Edit IP Settings]

The diagram illustrates the first two steps of the IP settings configuration. It starts with a screenshot of the 'Edit IP settings' dialog box where the 'Automatic (DHCP)' option is selected in the dropdown menu. A red box highlights this dropdown. An arrow points to the next screenshot, where the 'Manual' option is now selected in the dropdown menu, also highlighted with a red box. Below the dropdown, the 'IPv4' and 'IPv6' toggle switches are shown, both currently set to 'Off'. At the bottom of the dialog are 'Save' and 'Cancel' buttons.

Change [IPv4] from [Off] to [On], enter
[192.168.250.200] for the IP address and
[255.255.255.0] for the subnet mask, and
click [Save] (this is the IP address setting
on the PC side).

The diagram illustrates the final steps of the IP settings configuration. It starts with a screenshot of the 'Edit IP settings' dialog box where the 'Manual' option is selected in the dropdown menu. An arrow points to the next screenshot, where the 'IPv4' toggle switch is now turned 'On' (highlighted with a red box). Below the toggle, the 'IP address' field is set to '192.168.250.200' and the 'Subnet mask' field is set to '255.255.255.0', both fields highlighted with a red box. Below these fields are 'Save' and 'Cancel' buttons. A final arrow points to a third screenshot where the 'Save' button is highlighted with a red box, indicating the final step of the configuration.

2. Connection Method – Checking Communication Unit Connection (1/2) –

The following operations can be performed without administrator privileges.

1. Connect a LAN cable and turn on the communication unit (ZP-EIP) power.

2. Enter “Command” in the search bar and click [Command Prompt].



2. Connection Method – Checking Communication Unit Connection (2/2) –

The following operations can be performed without administrator privileges.

3. Enter **[ping 192.168.250.1]** (the IP address of the communication unit (ZP-EIP)) after **[C:¥Users¥****>]** and then press Enter.

* XXX: Enter the IP address of the communication unit. The default IP address of the communication unit is “192.168.250.1.”

```
Command Prompt
Microsoft Windows [Version 10.0.22000.2416]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Omron>ping 192.168.250.1
```

4. If you receive a response to the ping sent, the connection with the communication unit is normal.

```
Command Prompt
Microsoft Windows [Version 10.0.22000.2416]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Omron>ping 192.168.250.1

Pinging 192.168.250.1 with 32 bytes of data:
Reply from 192.168.250.1: bytes=32 time<1ms TTL=128
Reply from 192.168.250.1: bytes=32 time<1ms TTL=128
Reply from 192.168.250.1: bytes=32 time<1ms TTL=128
Reply from 192.168.250.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.250.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Omron>
```

* When a connection cannot be established, the message “Request timed out.” appears.
(The figure below is an example of mistakenly entering 2 instead of 1 as the last digit.)

In a case such as this, return to the procedure above and check whether the wrong IP address was entered, whether the IP address of the communication unit is wrong, whether the device power is turned on, and whether there is a LAN cable connecting the PC and communication unit.

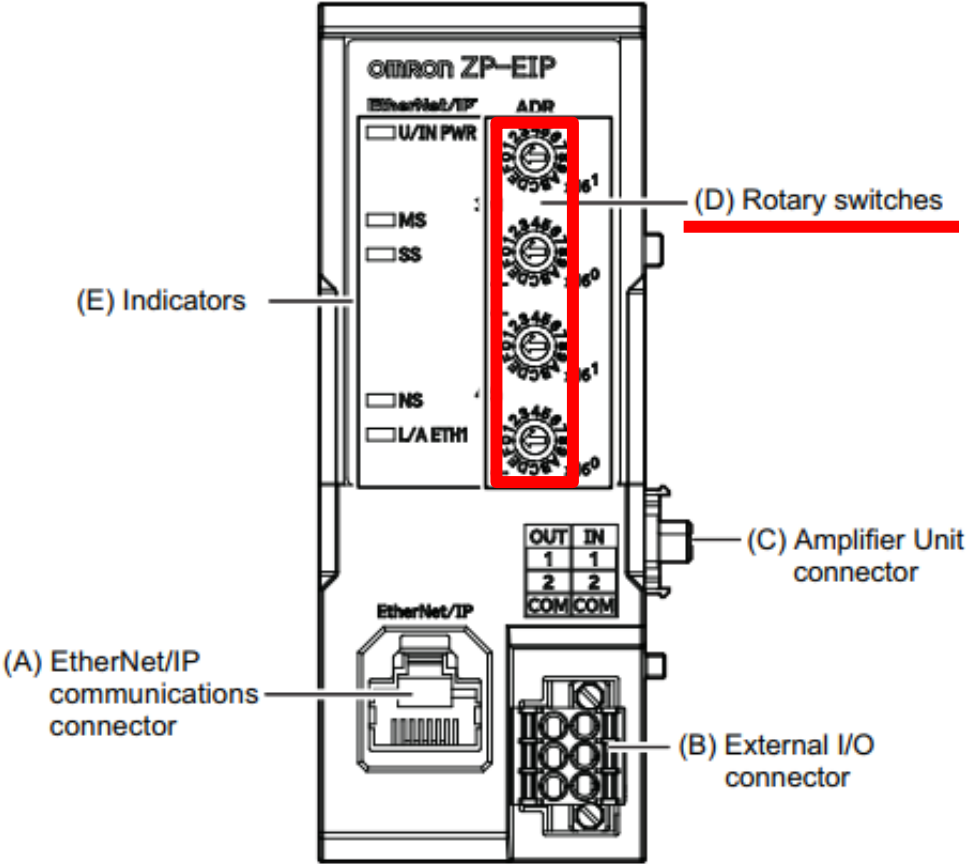
```
C:\Users\Omron>ping 192.168.250.2

Pinging 192.168.250.2 with 32 bytes of data:
Reply from 192.168.250.51: Destination host unreachable.
Reply from 192.168.250.51: Destination host unreachable.
Reply from 192.168.250.51: Destination host unreachable.
Reply from 192.168.250.51: Destination host unreachable.
```

2. IP Address Setting of Communication Unit

Change the rotary switches below to change the IP address of the communication unit.

All the rotary switches are set to 0 when the communication unit is shipped from the factory and the IP address of the unit is automatically set to “192.168.250.1.”
As the IP address is set by the hardware switches if a rotary switch for the 4th octet is changed from “00,” the 3rd octet rotary switches must also be changed.



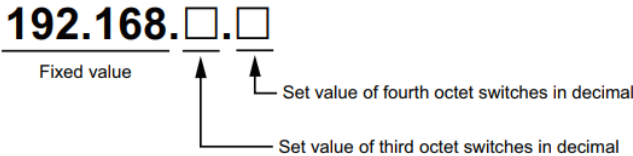
■ Communication Unit User’s Manual

There are several ways to set the IP address of the Communications Unit.
Use the rotary switches on the front panel of the Unit to specify the setting method for the IP address.

Set value (hex)		Setting Method	Reference
Third octet	Fourth octet		
00 to FF	00	Directly Setting the IP Address with the Network Configurator	Directly Setting the IP Address with the Network Configurator on page 3-9
		Getting the IP Address from the BOOTP Server with the Network Configurator	Getting the IP Address from the BOOTP Server with the Network Configurator on page 3-11
		Getting the IP Address from the DHCP Server with the Network Configurator	Getting the IP Address from the DHCP Server with the Network Configurator on page 3-13
00 to FF	01 to FE	Directly Setting the IP Address with Hardware Switches	Directly Setting the IP Address with Hardware Switches on page 3-15

● Directly Setting the IP Address with Hardware Switches

The following describes how to set the IP Address directly with hardware switches.
Use the third and fourth octet switches to directly set the IP address.
The relationship between the switch settings and the set IP address are as follows.

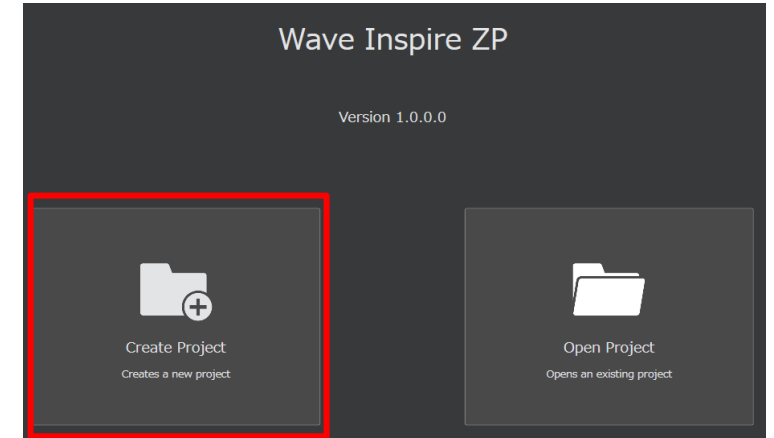


3. Startup Screen / Project Creation

Startup Screen (1/2)

When the tool starts up, the screen on the right appears. This tool is designed to be connected to a displacement sensor system, so you will need to prepare the system in advance.

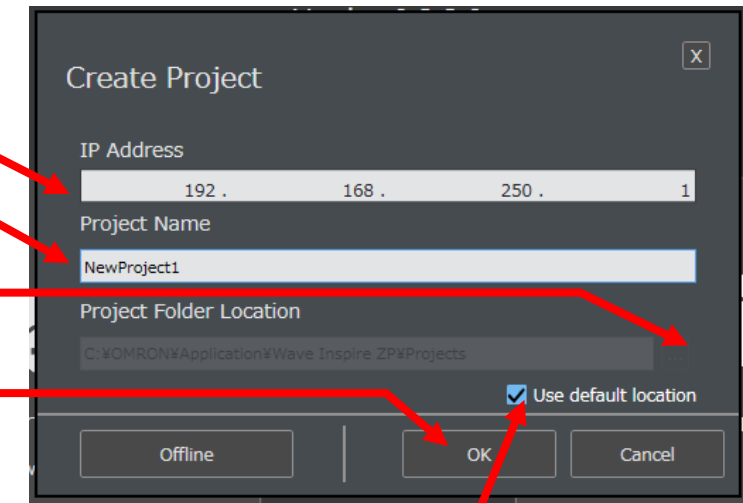
- Create Project
 - ... Create a new project.
This is used to connect to a ZP system and create a project file matching the system configuration of the ZP system.
- Open Project
 - ... Open a previously saved project file.



<Creating a new project>

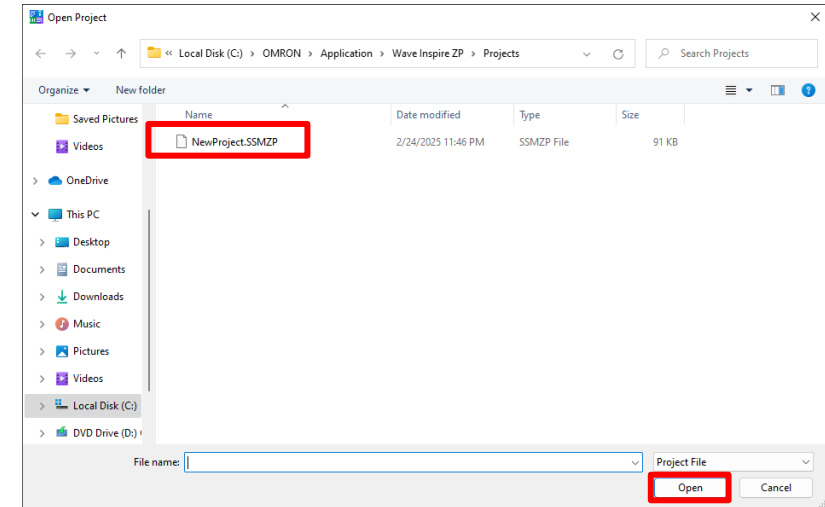
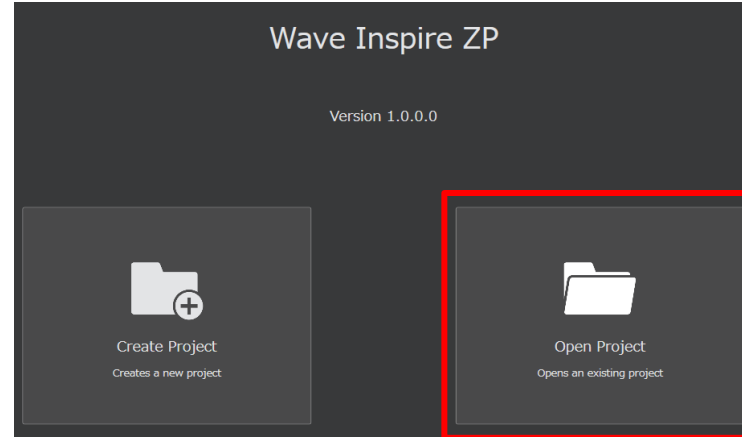
1. Click the [Create Project] button.
2. Enter the IP address.
3. Enter the project name.
4. Select the save destination.
Click the [...] button to launch Explorer and select a folder.
5. Click the [OK] button.

A project file is created. The PC is connected to the communication unit with the configured information.



If this check box is selected (☑), the default folder specified during installation is referenced.

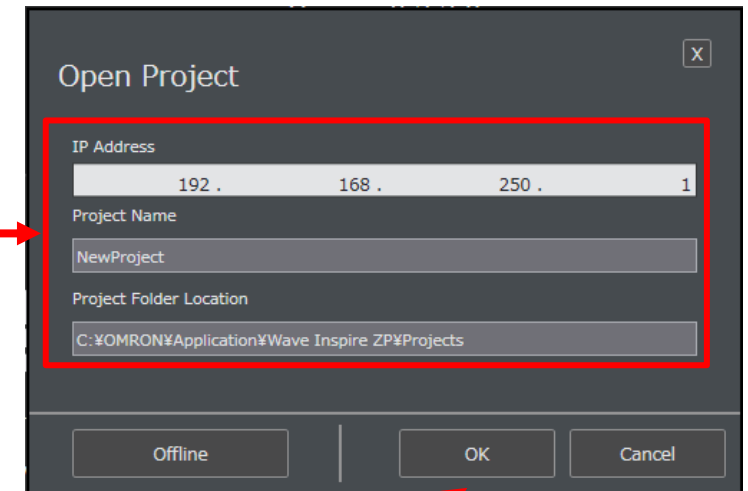
Startup Screen (2/2)



<Opening an existing project>

1. Click the [Open Project] button.
2. Select and open the project file in Explorer.
3. The IP address, project name, and save destination are filled in.
4. Click the [OK] button.

A project file is created. The PC is connected to the communication unit with the configured information.



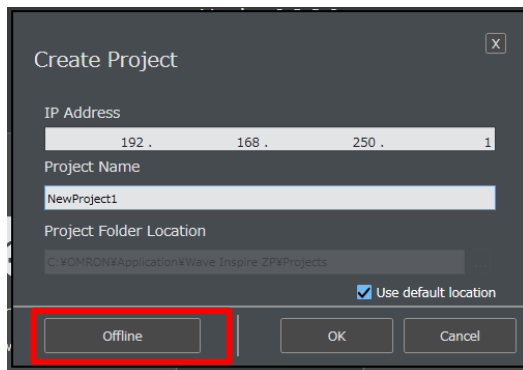
Startup Screen – Offline Startup

This PC tool can be started even when it is not connected to a sensor (offline state).
The following operations are possible even if no sensor is available.

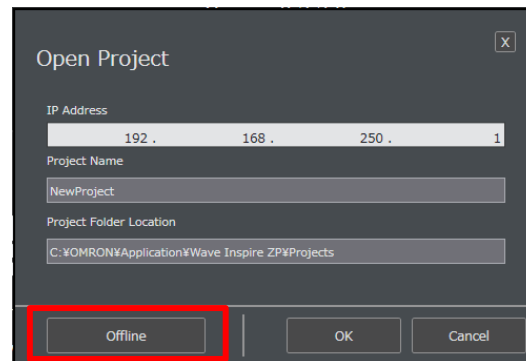
- Checking and editing the sensor settings stored in the project
- Checking and editing settings related to graphic display with the PC tool

As no sensor is connected in the offline state, it is not possible to transition to the “Sensor States” screen.

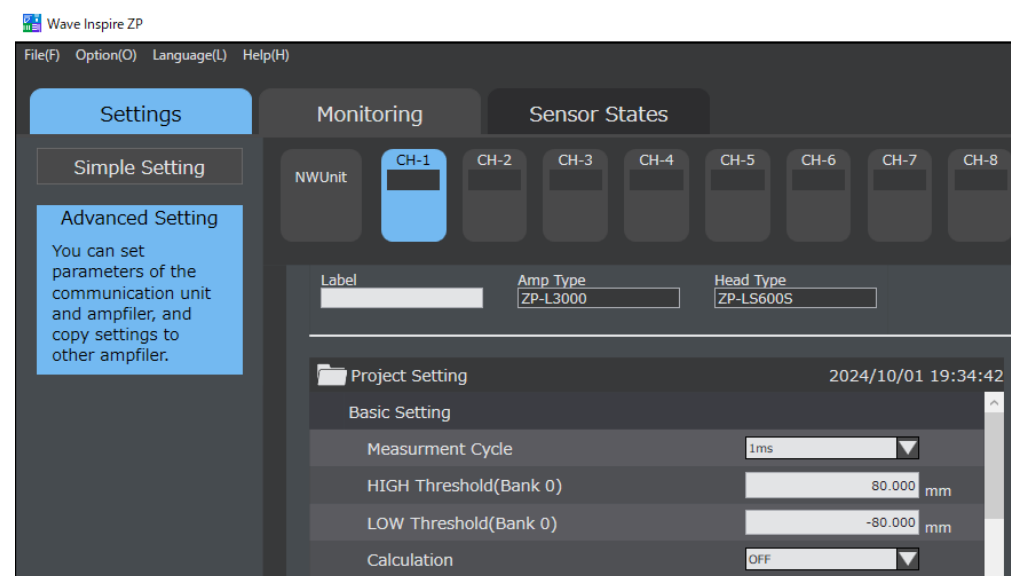
<Creating a new project>



<Opening an existing project>



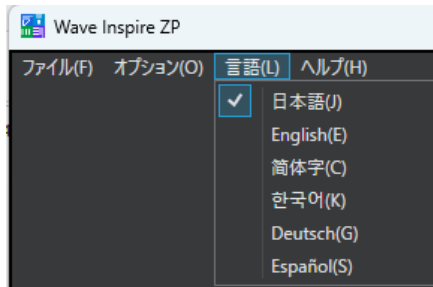
<Screens in offline state>



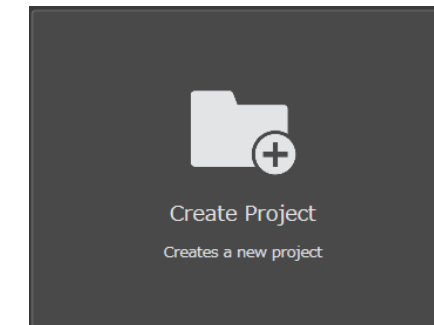
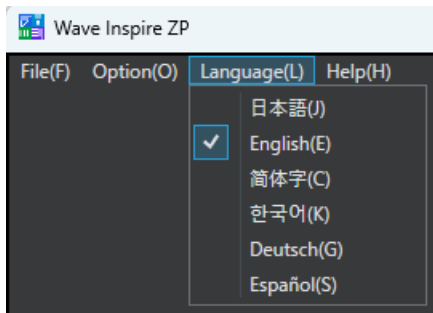
Startup Screen – Language Switching

Changing [Language] in the menu allows you to change the display language of the PC tool without restarting. Japanese, English, Simplified Chinese, Korean, German, and Spanish are supported.

Japanese



English



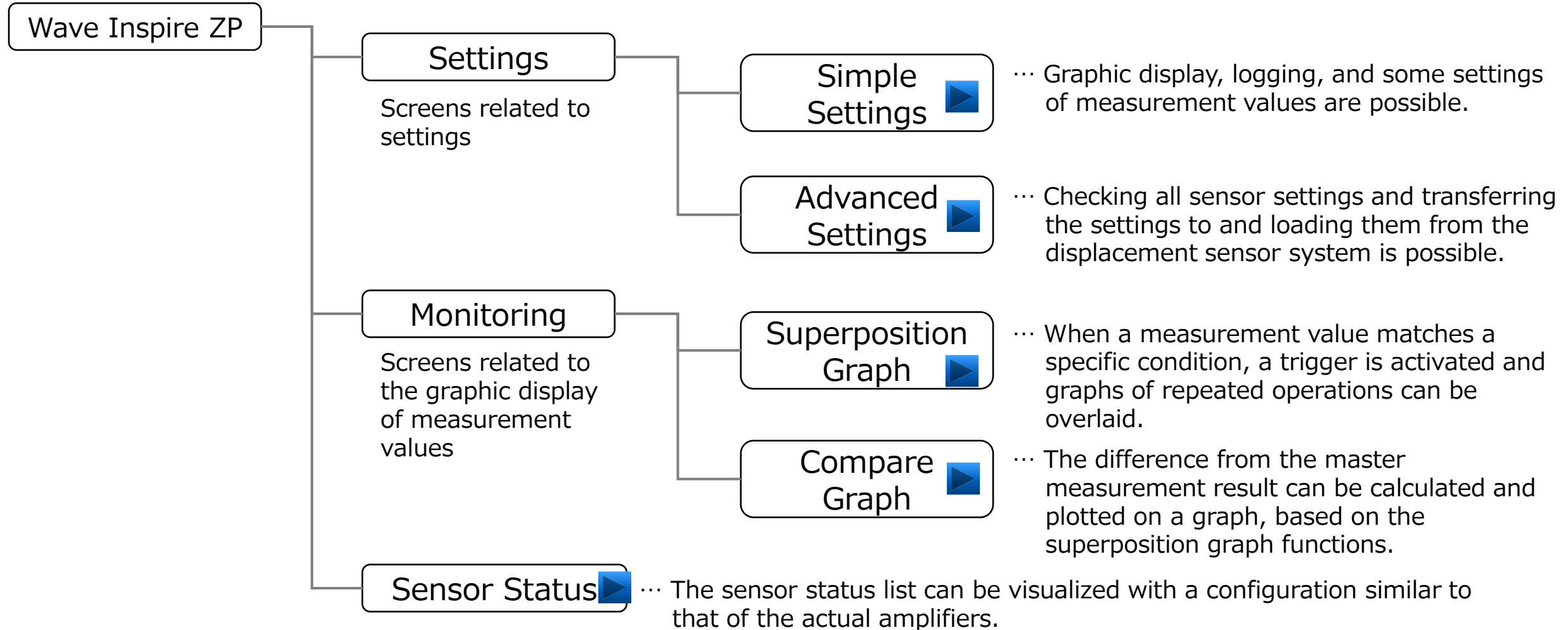
Simplified Chinese



4. Overview of Functions and Screen Operations

Overall Screen Configuration

The overall screen configuration after connecting to a displacement sensor system is shown below. The details of each screen will be described later.



Settings – Simple Settings Screen: Key to Screen

This screen allows you to graphically display the measurement values and configure some items for the selected amplifier. The setting items are extracted only for those items whose setting values must be adjusted while observing the measurement values (items related to the judgment output threshold and the stability of the measurement values).

The screenshot shows the 'Simple Setting' screen with various controls and a graph. Annotations include:

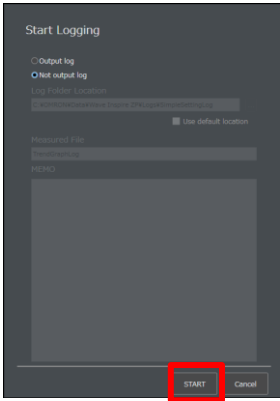
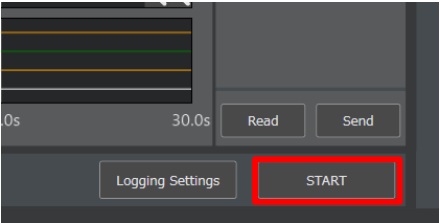
- Area to select the amplifier (CH-1 is currently selected.)**: Points to the top row of channel buttons (CH-1 to CH-16).
- Displays the difference between the two cursors at the left of the screen.**: Points to the 'Cursor Distance' field showing 168.313 mm.
- Select this check box (☑) to adjust the graph plotting range according to the maximum and minimum measurement values.**: Points to the 'Y-Axis Autoscale' checkbox.
- Graph zoom in/zoom out (vertical axis)**: Points to the vertical zoom controls on the left.
- Tabs to adjust the cursor positions**: Points to the 'H' and 'L' threshold tabs on the right.
- Area to graphically display the measurement values**: Points to the main graph area showing a sine wave.
- Area to graphically display the judgment outputs**: Points to the bottom section showing 'HIGH', 'PASS', 'LOW', and 'Timing' indicators.
- Buttons for specifying the High/Low judgment threshold values numerically**: Points to the 'H' and 'L' input fields on the right.
- Tabs to adjust the High/Low judgment threshold values**: Points to the 'H' and 'L' tabs on the right.
- Setting item area**: Points to the right-hand panel containing various settings.
- Graph zoom in/zoom out (horizontal axis)**: Points to the horizontal zoom controls on the right.
- Click the [Read] button to apply the settings from the displacement sensor system to the screen.**: Points to the 'Read' button at the bottom right.
- Click the [Send] button to apply the settings to the displacement sensor system.**: Points to the 'Send' button at the bottom right.

* Note that the displayed graph is reset if you switch to another screen, such as the Monitoring tab.

Settings – Simple Settings Screen: Graph Plotting and Logging Procedure

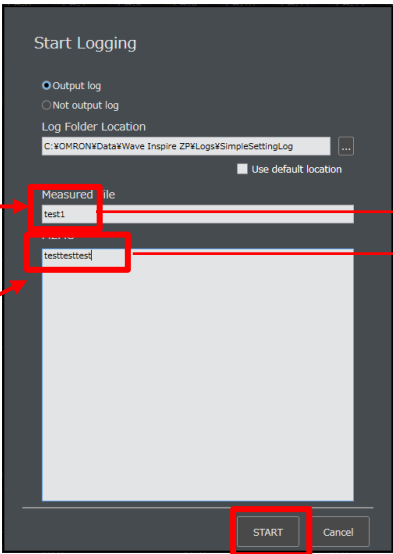
The following procedure enables graph plotting and logging.

<No csv output>



Start graph plotting

<csv output>



Start graph plotting and logging to csv

The characters entered in this text box are applied to the output file name.

The characters entered in this text box are applied to the memo field in the output file.

test1 20240414125047978.csv

	A	B	C	D	E	F	G
	ZP-L3050	0.208	ZP-LS300	0.208			
	memo	testtesttest					
Count	TimeStan	CH1	CH1Timin	CH1High	CH1Pass	CH1Low	
1	2024/04/	58.04211	0	1	0	0	
2	2024/04/	58.04236	0	1	0	0	

Index Time Measurement value Judgment output (0: OFF, 1: ON)

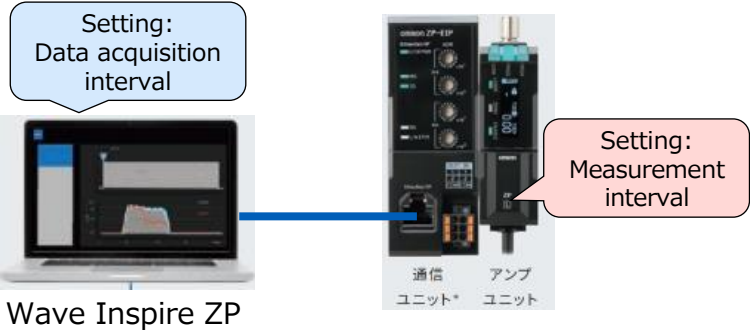
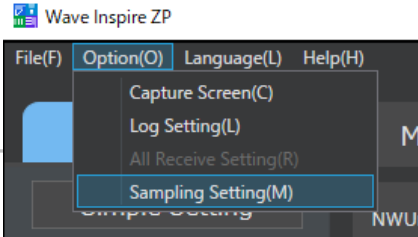
Settings – Overview of Logging Data

The logging data can be compressed by increasing the data acquisition interval during logging.

This can be set using [Option] - [Sampling Setting] as shown in the figure on the upper right.

The following occurs if the measurement interval of the sensor amplifier and the data acquisition interval of the tool differ.

■ Schematic representation of logging data



		PC tool settings – Data acquisition interval	
		1 ms	4 ms
Sensor amplifier settings – Measurement interval	1 ms	Each data value is different as the measurement values and data acquisition updates are aligned.	As the measured value is updated every 1 ms and the data acquisition takes 4 ms, the graph looks like this.
	4 ms	As the measured value is updated every 4 ms and the data acquisition takes 1 ms, the same data value appears four times consecutively.	Each data value is different as the measurement values and data acquisition updates are aligned.

Settings – Advanced Setting Screen

This screen allows you to check all settings for the selected amplifier and transfer the settings to and load them from the displacement sensor system.
You can immediately identify any points of difference that exist between the project and the displacement sensor system.

You can label each amplifier with any text string.

These alert icons notify you that the project settings differ from the displacement sensor system settings. This makes it easy to identify which amplifiers have differences in the settings.

Setting information stored in the project for the selected amplifier

The screenshot displays the 'Advanced Setting' screen for a selected amplifier (CH-1 measP1). The interface is divided into two main sections: 'Project Setting' on the left and 'Amp Setting' on the right. Both sections show a 'Basic Setting' table with parameters like Measurement Cycle, HIGH Threshold, LOW Threshold, Calculation, and Analog Output. The 'Project Setting' table has a red background for the 'LOW Threshold(Bank 0)' row, indicating a difference from the displacement sensor system settings. The 'Amp Setting' table also has a red background for the 'LOW Threshold(Bank 0)' row. Annotations include: a yellow circle around the 'measP1' label; a yellow circle around the 'LOW Threshold(Bank 0)' row in both tables; a yellow circle around the 'Send' button; a yellow circle around the 'Execute' button; and a yellow circle around the 'Reload' button (a circular arrow icon) in the top right of the 'Amp Setting' panel. A yellow bracket at the bottom groups the 'Project Setting' and 'Amp Setting' panels under the labels 'Setting information stored in the project for the selected amplifier' and 'Displacement sensor system setting information for the selected amplifier' respectively.

Displacement sensor system setting information for the selected amplifier

Reload the displacement sensor system settings. (Use this after changing the settings using the amplifier buttons after connecting the PC tool.)

Transfer the settings from the project to the displacement sensor system.

Load the settings from the displacement sensor system to the project.

Use this button to perform operations such as initialization and zero reset of the displacement sensor system.

* This is the execution button for the displacement sensor system. It does not apply to the [Send] or [Receive] buttons.

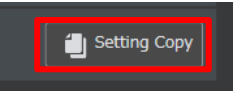
Settings – Advanced Setting Screen:

Steps to Duplicate Settings between Amplifiers

When multiple amplifiers are cascaded together, settings from one amplifier can be duplicated to another by following the steps below.

Select the amplifier to duplicate the settings to and click the [Copy & Send Sensor] button.

* When the settings are duplicated, the settings are immediately applied to the displacement sensor system.



With the amplifier you want to duplicate the settings selected, click the [Setting Copy] button at the bottom right of the screen.

A screenshot of the "Copy Settings" dialog box. It has a "Copy from" section with fields for "Channel No" (CH-2) and "Label". Below this are fields for "AMP Type" (ZP-L3510) and "Head Type" (ZP-LS300S). A "Copy to" section shows a grid of 16 channels (CH-1 to CH-16). CH-3 and CH-4 are selected with blue checkmarks. At the bottom right, there is a "Copy & Send Sensor" button highlighted with a red border and a "Cancel" button. A yellow arrow points to the "Copy & Send Sensor" button.

* See the next page to see the conditions that determine whether the settings can be duplicated.

Click the [OK] button.

A screenshot of the "Copy Settings" dialog box. It shows "Copy from CH-2" and "Copy to" with a list of channels (CH-3, CH-4). The "OK" button at the bottom is highlighted with a red rectangular border.A screenshot of the "Copy Settings" completion screen. It displays "Copy Settings completed." followed by a blue progress bar at 100%. There is an "OK" button at the bottom.

Duplication complete

Settings – Advanced Setting Screen:

Possibility to Duplicate Settings between Amplifiers

The setting duplication function is subject to the following restrictions because the source and destination setting items must match.

- Amplifier: The models must match exactly
- Head: The detection distance ranges must match
(The underlined part of the format to the right must match. ZP-LS100S)

Examples of cases are shown below.

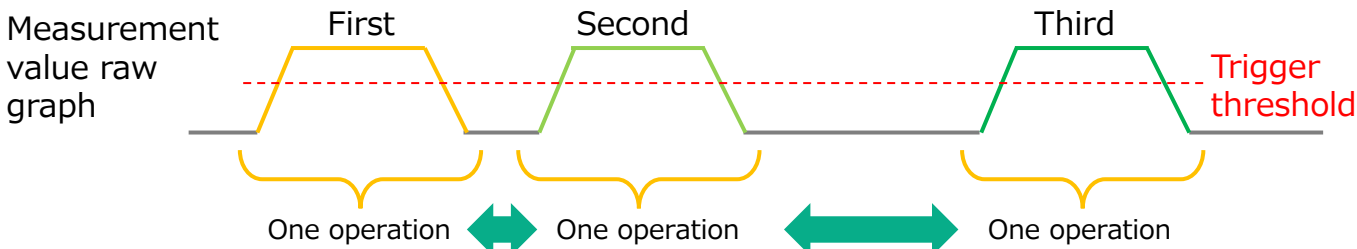
Case	Duplicated from			Duplicated to			Possibility of duplication
	Ch.	Amplifier	Head	Ch.	Amplifier	Head	
Example 1	1	ZP-L3000 (parent)	ZP-LS100S	2	ZP-L3500 (child)	ZP-LS100S	NO
Example 2	2	ZP-L3500 (child)	ZP-LS100S	3 to 16	ZP-L3500 (child)	ZP-LS100S	YES
Example 3	2	ZP-L3500 (child)	ZP-LS100S	3	ZP-L3500 (child)	ZP-LS600S	NO
Example 4	2	ZP-L3500 (child)	ZP-LS100S	3	ZP-L3500 (child)	ZP-LS100L	YES
Example 4	2	ZP-L3500 (child)	ZP-LS100S	3	ZP-L3500 (child)	ZP-LS100SC (Laser class 1)	YES
Example 5	2	ZP-L3500 (child)	ZP-LS100S	3	ZP-L3590 (child, no I/O)	ZP-LS100S	NO

Monitoring – Superposition Graph: Aim of Function

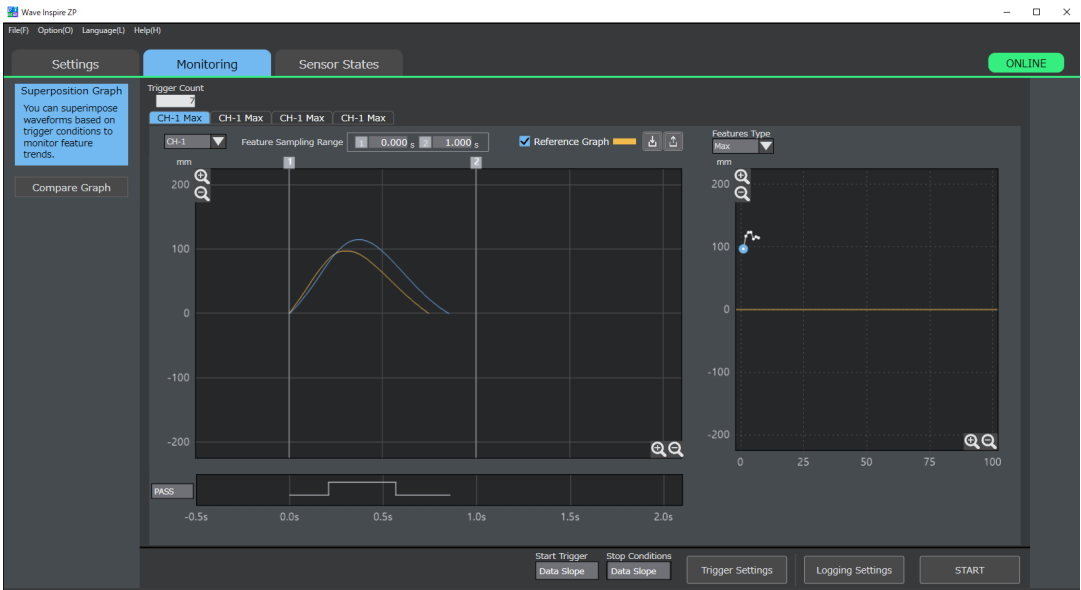
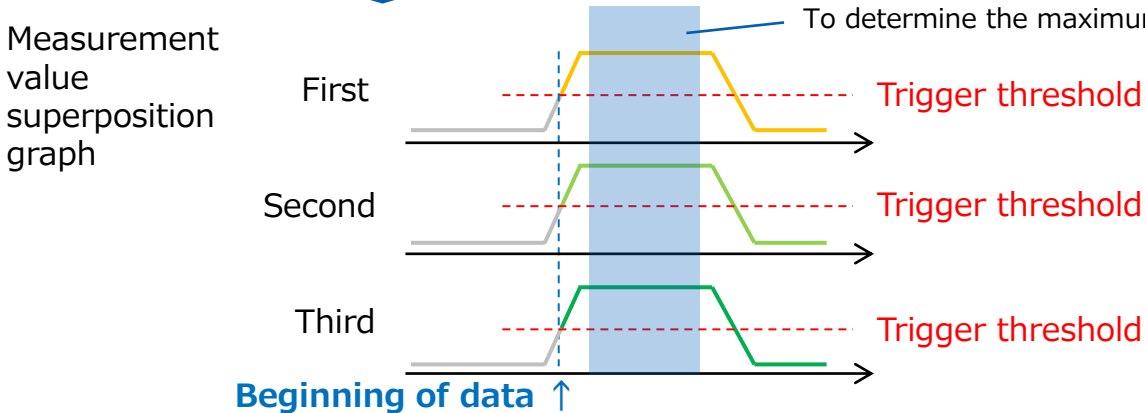
By setting trigger conditions for specific repetitive operations, it is possible to cue each operation, align the timing, and acquire and plot the measurement values.

A schematic representation is shown below.

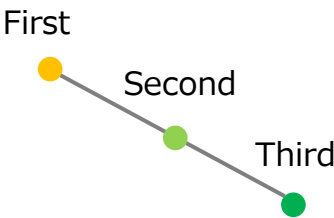
Example: Each operation is almost identical but the timing between the operations varies.
We want to extract a feature value (for example, the maximum value) for each of these operations.



Cued when the threshold is exceeded as a trigger condition



To visualize the transition of maximum values for each operation



Plot the maximum values (feature values)

Monitoring – Superposition Graph: Key to Screen

By setting trigger conditions for specific repetitive operations, it is possible to cue each operation, align the timing, and acquire and plot the measurement values.

Count of the number of times the trigger is activated.

A total of four feature values can be set.

Select the measurement amplifier.

Use the cursor to specify the range to calculate the feature values.

Specify the feature value to calculate. (Select from the following five types.)

Max

Min

Ave

P-p

Sum

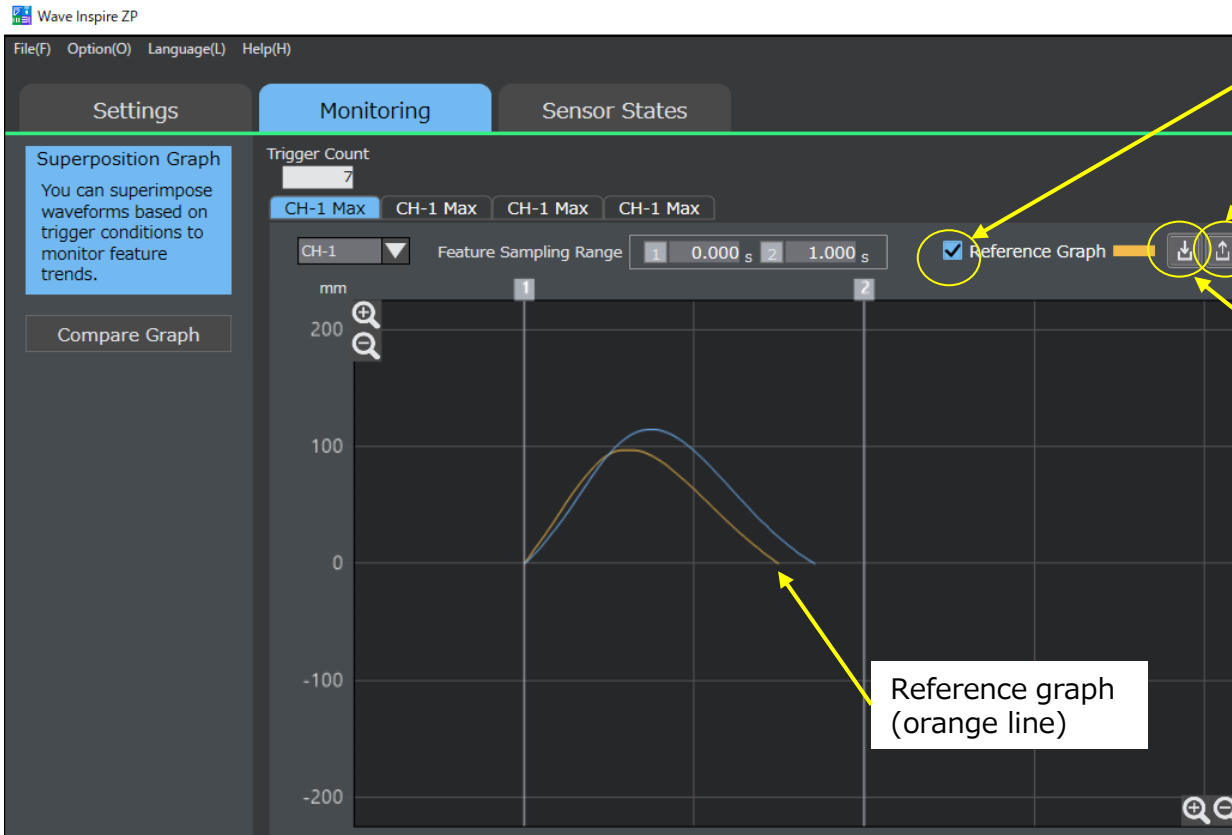
* Note that the displayed graph is reset if you switch to another screen, such as the Monitoring tab.

Monitoring – Superposition Graph: Key to Reference Graph

The graph of one measurement is plotted as an orange line as a reference.

It can be displayed as an example of ideal device operation as a reference, for example.

If no data is specified for the reference graph, the measurement graph for the first trigger is plotted as the reference graph.



The reference graph is displayed if this check box is selected (☑).

(1) Export the reference graph
Exports the currently plotted reference graph (orange line graph) as data. When the Explorer window appears, select the save destination.

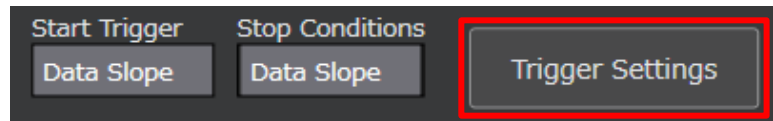
(2) Load the reference graph
The data exported at step (1) is loaded and displayed as the reference graph (orange line). When the Explorer window appears, select where to load the data from.

* Note that only data exported at step (1) can be loaded.
Provided the format matches, you can edit the data from step (1).

Monitoring – Superposition Graph: Setting Trigger Conditions (1/2)

The trigger conditions can be set as follows.
Some of these are explained on the next page.

Click the [Trigger Settings] button at the bottom right of the screen.

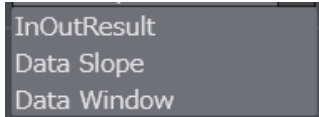


Current trigger settings



Select the amplifier to be used as the trigger condition.

Select from the following three options. The details are shown on the next page.

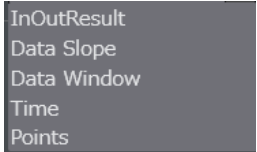


Set the start trigger condition to start graph plotting.

This can be set to plot the graph only when the trigger is activated multiple times. For example, if set to 2: The data is saved after every two triggers and the graph is plotted after every second one.

Set the end trigger condition to end graph plotting.

Select from the following five options. The details are shown on the next page.

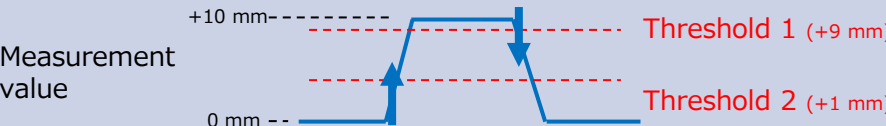


Settings complete.
The procedure for starting measurement/logging is the same as on the Simple Settings screen.



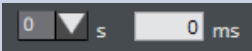
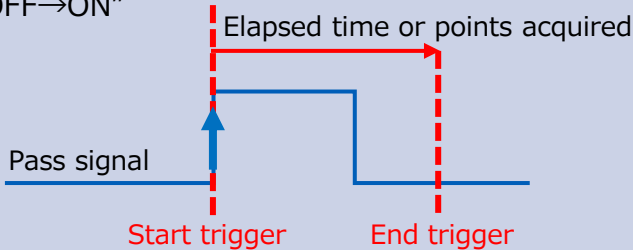
Monitoring – Superposition Graph: Setting Trigger Conditions (2/2)

The following three start trigger conditions are available.

Trigger condition and explanation	Data used	Details of setting	Threshold	Schematic representation
I/O result … A trigger is activated when a judgment output turns ON/OFF.	High/Pass/ Low/etc.	OFF→ON or ON→OFF	None	Trigger timing when set to “Pass signal” or “OFF→ON” 
Data slope … A single threshold is set and a trigger is activated when the measurement value crosses the threshold (rising or falling edge).	Measurement value	Rising or falling edge	One setting	Trigger timing when set to “Rising edge” 
Data window … Two threshold values are set and a trigger is activated when the measurement value enters or leaves that range.	Measurement value	IN or OUT	Two settings	Trigger timing when set to “IN” (In this example, the trigger is activated twice.) 

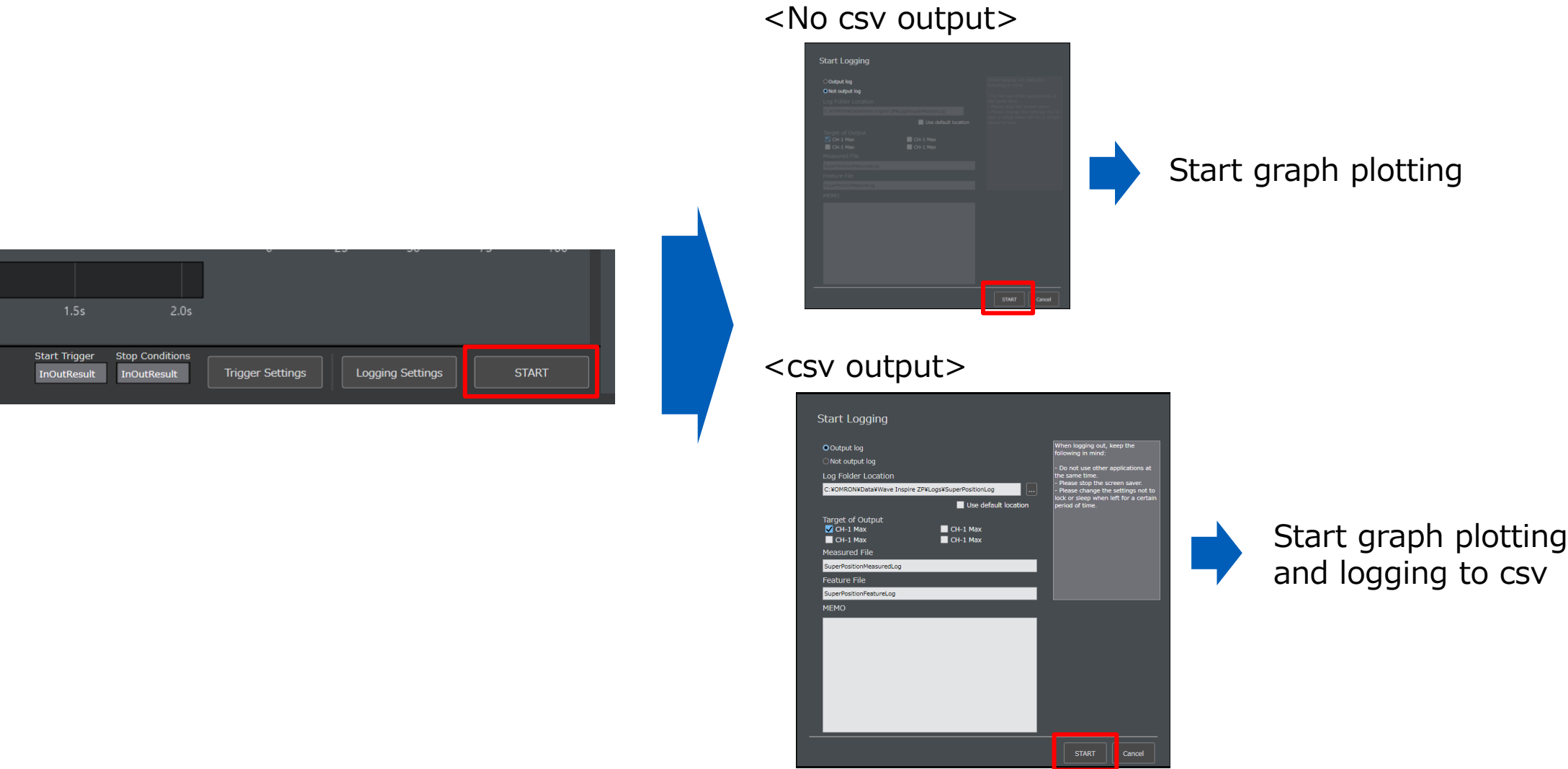
In addition to the three conditions above, the following two are added as end trigger conditions.

* If no end trigger is detected for 60 seconds after a start trigger is detected, an end trigger is forcibly detected.

Trigger condition and explanation	Data used	Details of setting	Threshold	Schematic representation
Elapsed time … After a start trigger is detected, the end trigger is activated when the set time has elapsed.	-	Time 	-	End trigger timing when the start trigger is set to “Pass signal” or “OFF→ON” 
Points acquired … After a start trigger is detected, the end trigger is activated when the set number of points has been acquired.	-	Points 	-	

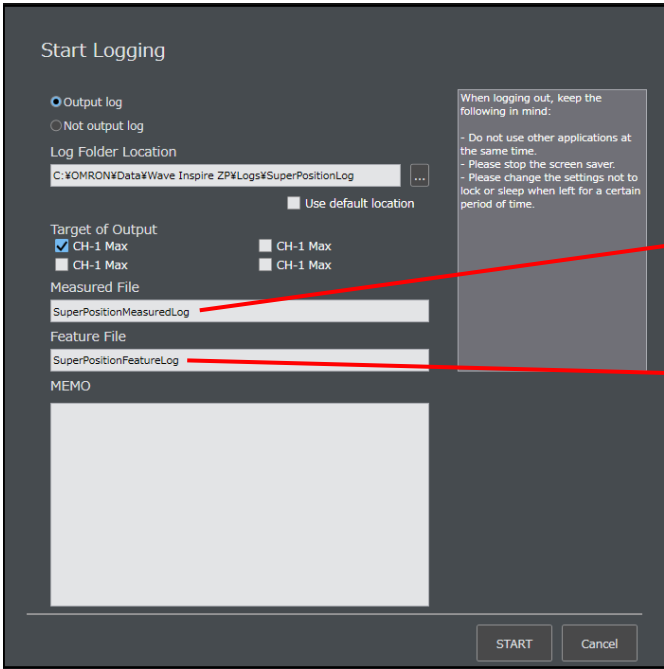
Monitoring – Superposition Graph: Graph Plotting and Logging Procedure

The following procedure enables graph plotting and logging.



Monitoring – Superposition Graph: Logging Data

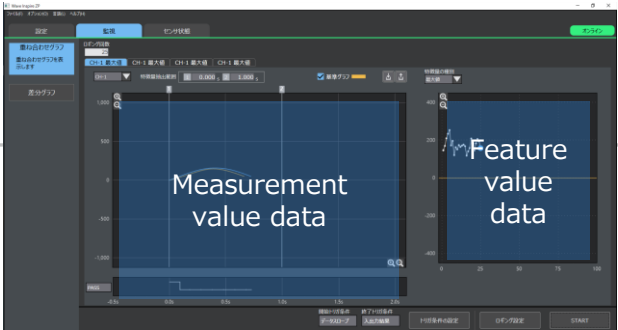
The logging data configuration is as follows.



名前	更新日時
SuperPositionMeasuredLog_Tab1_CH1_202406281427...	2024/06/28 14:27
SuperPositionMeasuredLog_Tab2_CH1_202406281427...	2024/06/28 14:27
SuperPositionMeasuredLog_Tab3_CH1_202406281427...	2024/06/28 14:27
SuperPositionMeasuredLog_Tab4_CH1_202406281427...	2024/06/28 14:27
SuperPositionFeatureLog_Tab1_CH1_20240628142706...	2024/06/28 14:27
SuperPositionFeatureLog_Tab2_CH1_20240628142706...	2024/06/28 14:27
SuperPositionFeatureLog_Tab3_CH1_20240628142706...	2024/06/28 14:27
SuperPositionFeatureLog_Tab4_CH1_20240628142706...	2024/06/28 14:27

Measurement value data
This is the measurement value data cued by the trigger.
(Data in the graph at the left of the screen above)
As four data can be set on the tab, four files are output.

Feature value data
This is the feature value data calculated using the measurement value data. (Data in the graph at the left of the screen above)
As four data can be set on the tab, four files are output.



■ SuperPositionMeasuredLog ~.csv

	A	B	C	D	E	F	G	H
1	ZP-L3000	ZP-LS05C	CH1					
2	memo	testtesttest						
3	StartTrig	InOutRes	PASS	OFF→ON				
4	StopTrig	Time	0h 0m 1s 0ms					
5	StartPoint	0						
6	EndPoint	1000						
7	Cycle	TimeStar	Start					
8	CriteriaData		-2	-1.999	-1.998	-1.997	-1.996	-1.995
9	*1	1	47:10.9	-2	-1.999	-1.998	-1.997	-1.996
10		2	47:20.8	-2	-1.999	-1.998	-1.997	-1.996
11		3	47:30.7	-2	-1.999	-1.998	-1.997	-1.996

Index (Number of triggers) Time Measurement value data

■ SuperPositionFeatureLog ~.csv

	A	B	C	D	E	F	G
1	ZP-L3000	ZP-LS05C	CH1				
2	memo	testtesttest					
3	Cycle	TimeStar	MAX	MIN	AVE	P-p	SUM
4		1	47:10.9	-1.001	-2	-1.5005	0.999
5		2	47:20.8	-1.001	-2	-1.5005	0.999
6		3	47:30.7	-1.001	-2	-1.5005	0.999
7		4	47:40.9	-1.001	-2	-1.5005	0.999
8		5	47:50.8	-1.001	-2	-1.5005	0.999

Index (Number of triggers) Time Feature value data

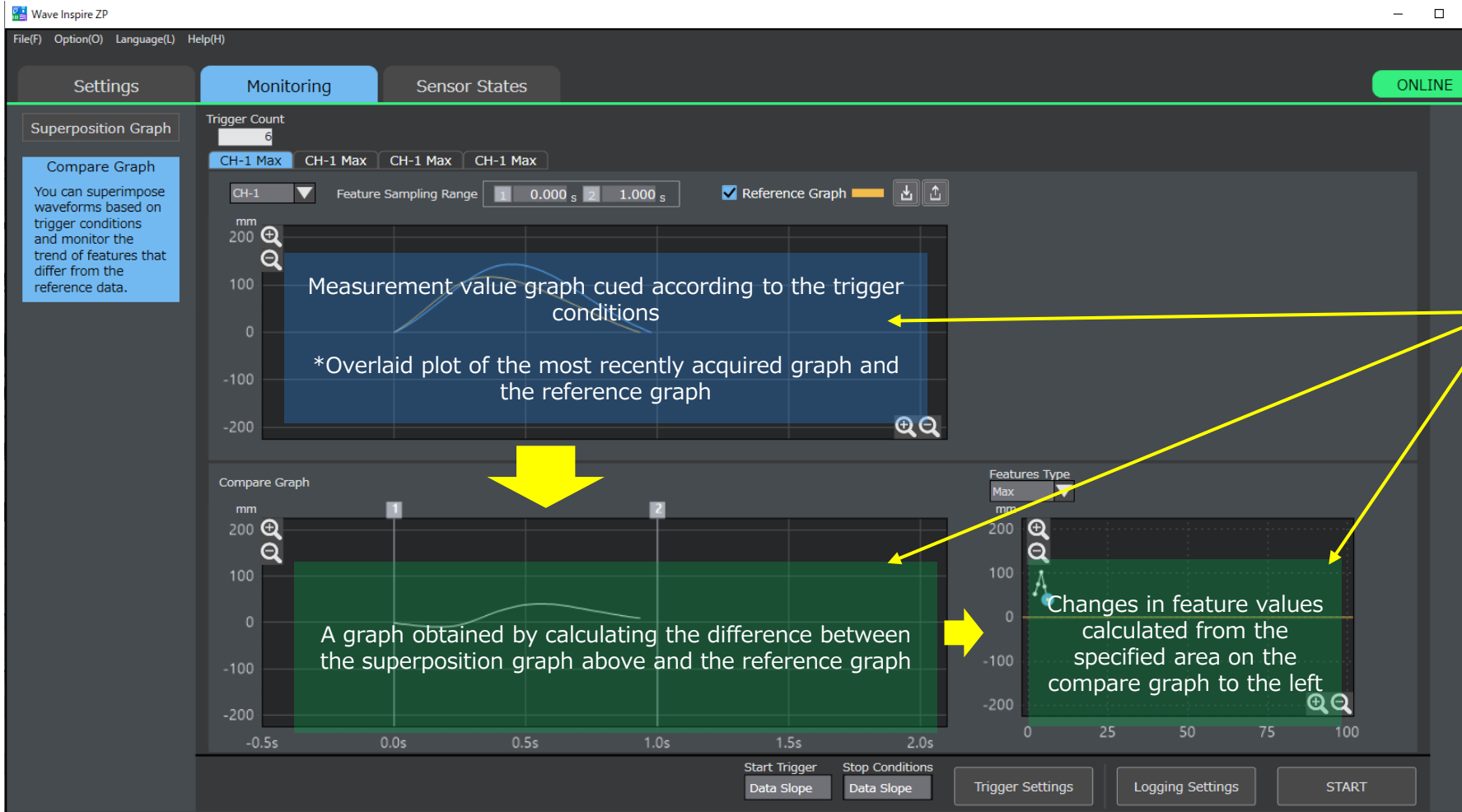
*1: Each time a trigger is activated, data is added in the direction of the arrow.

Monitoring – Compare Graph: Key to Screen

This screen is an expanded version of the superposition graph described in the previous pages.

For the measurement values cued on the superposition graph, the difference from the reference graph is calculated and plotted on the bottom-left screen.

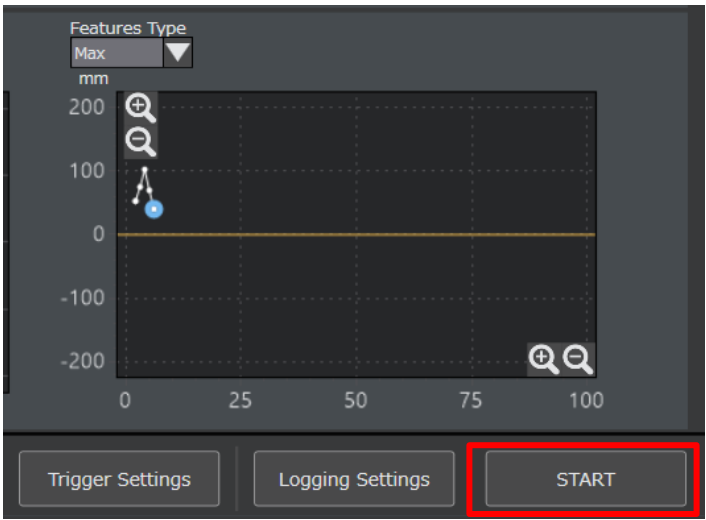
All button functions not related to comparison processing behave the same as on the superposition graph.



* Note that the displayed graph is reset if you switch to another screen, such as the Monitoring tab.

Monitoring – Compare Graph: Graph Plotting and Logging Procedure

The following procedure enables graph plotting and logging.



<No csv output>



Start graph plotting

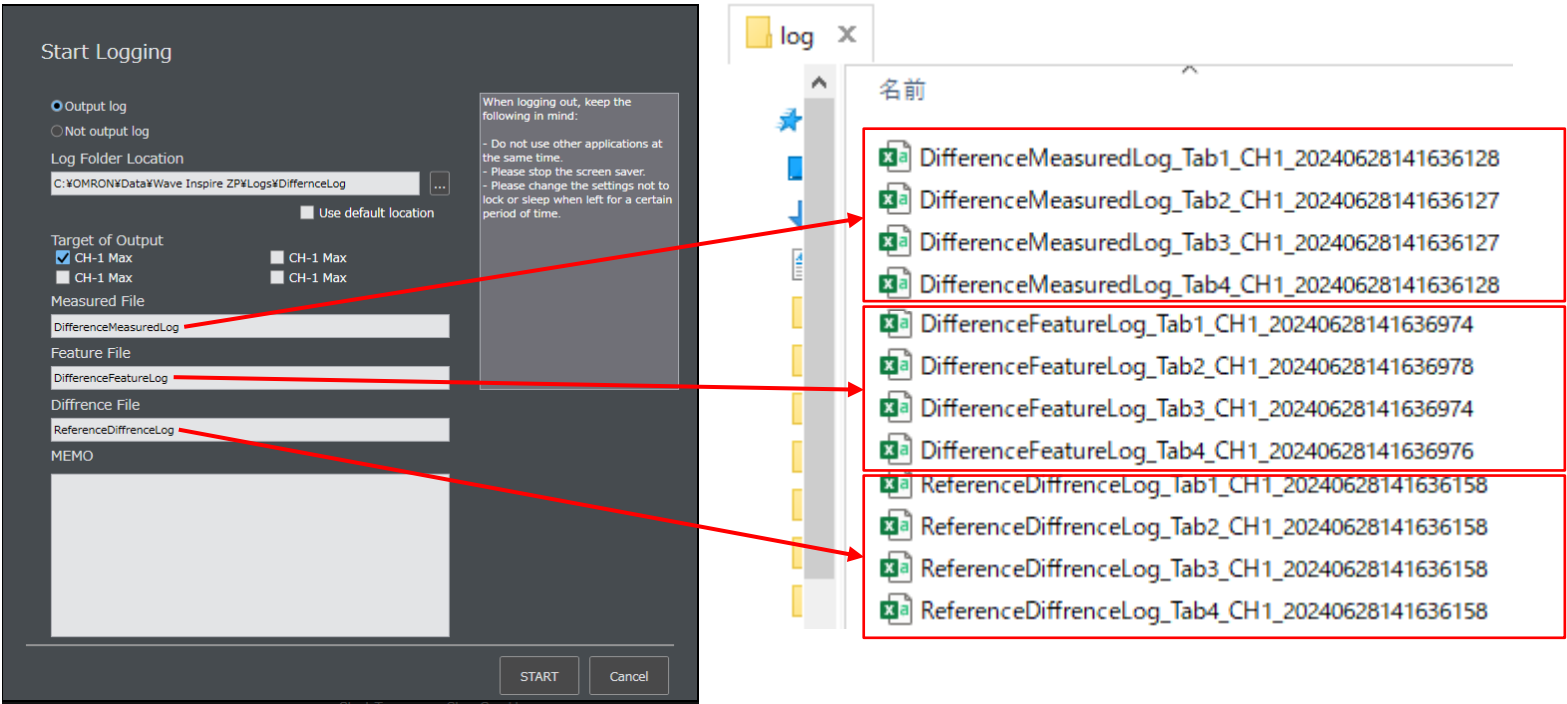
<csv output>



Start graph plotting and logging to csv

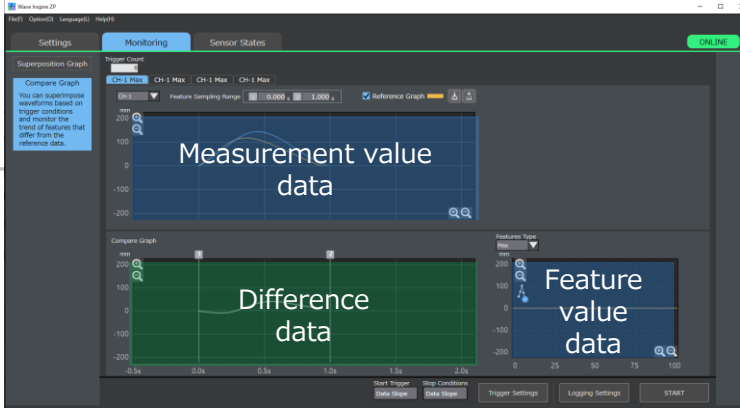
Monitoring – Compare Graph: Logging Data

The logging data configuration is as follows.



- DifferenceMeasuredLog ~.csv
- DifferenceFeatureLog ~.csv

Omitted here as it has the same configuration as the superposition graph.



- Measurement value data**
This is the measurement value data cued by the trigger.
(Data in the graph at the top-left of the screen above)
As four data can be set on the tab, four files are output.
- Feature value data**
This is the feature value data calculated using the difference data.
(Data in the graph at the right of the screen above)
As four data can be set on the tab, four files are output.
- Difference data**
This is the measurement value data calculated from the difference between the measurement value data acquired by the trigger and the measurement value data of the reference graph.
(Data in the graph at the bottom-left of the screen above)
As four data can be set on the tab, four files are output.

■ ReferenceDifferenceLog ~.csv

	A	B	C	D	E	F	G	H
1	GP-L3010	GP-LS6000 CH1						
2	memo	testtesttest						
3	StartTrigger	データスロープ 立ち上がり		0				
4	StopTrigger	データスロープ 立ち下がり		0				
5	StartPoint	0						
6	EndPoint	1000						
7	Cycle	TimeStamp Start						
8	1	16:34.9						
9	2	16:36.0	0.21068	0.12493	0.03712	-0.05043	-0.1385	-0.2278
10	*1	3	16:37.2	-21474.5	-21473.8	-21473.1	-21472.5	-21471.8
11		4	16:38.4	-21474.5	-21473.8	-21473.1	-21472.5	-21471.8
12		5	16:40.0	-0.12007	-0.0681	-0.01843	0.03149	0.07885

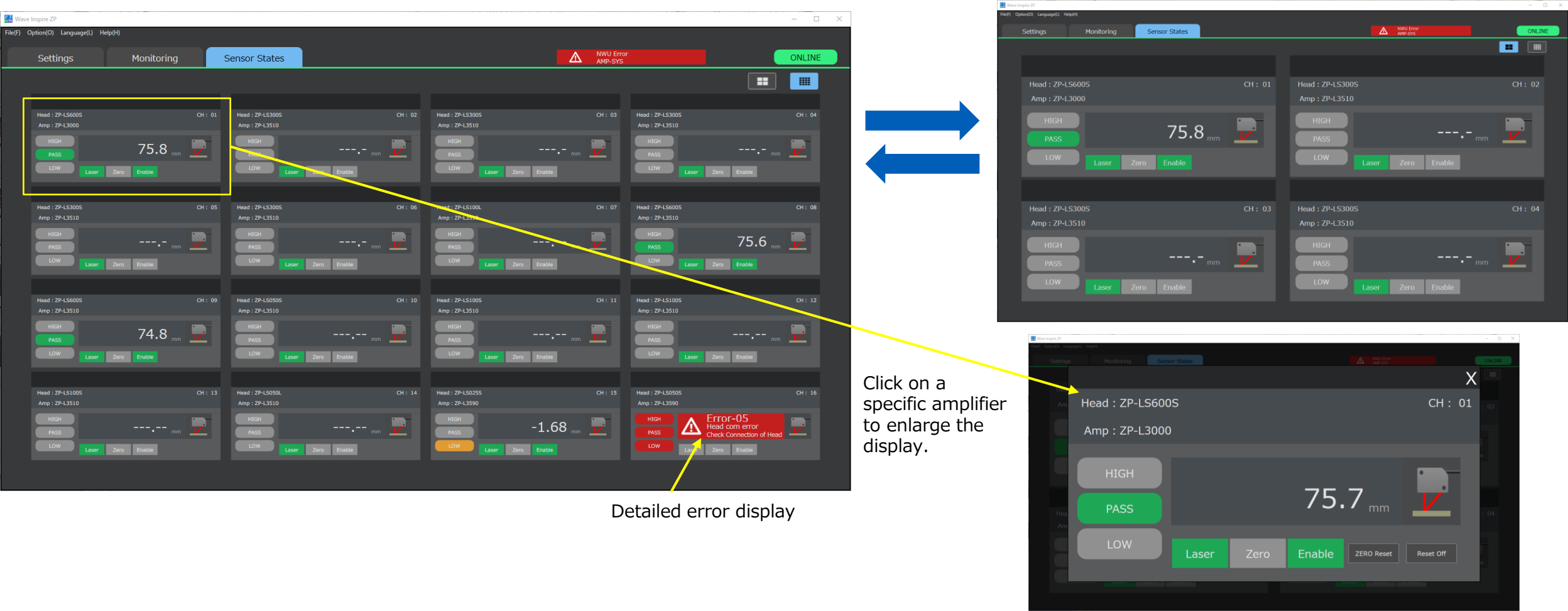
Index (Number of triggers) Time Difference data

*1: Each time a trigger is activated, data is added in the direction of the arrow.

Sensor Status: Key to Screen

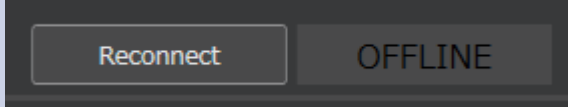
This screen shows an overview of the measurement values and error status of the sensors connected to the ZP system.

* The sensor icon changes according to the type of sensor.



5 . Cautions and Troubleshooting

Troubleshooting

No	Contents	Correction	References
1	The LAN cable was disconnected while the PC tool was connected.	After the error notification, the following button will appear in the upper right corner of the screen. Reconnect the LAN cable and press the “Reconnect” button to recover.  ※ If the connection is lost while drawing a graph on the Easy Setup screen, etc., the connection cannot be reconnected for 60 seconds; wait 60 seconds and then reconnect.	—
2	Logging is no longer available.	It is assumed that there is insufficient space in the storage location. Please delete unnecessary files to secure the space.	—
3	Cannot read out data as a reference graph for the superposition graph.	Reference graphs can only be read if the data format is the same screen output.	
4	It takes time to switch screens and switch amplifier CHs.	Since the memory space in the PC may be insufficient, please close other applications and then restart Wave Inspire ZP.	—

No	Contents	Correction	References
1	Is it possible to make settings using the buttons on the amplifier while the PC tool is connected? If so, which settings are reflected, the PC tool settings or the amplifier button settings?	Settings can be made using the amplifier buttons even when the PC tool is connected. When both settings are made with the amplifier button and from the PC tool, the settings made later will be reflected.	—
2	Does the buffering storage interval of the communication unit affect the logging data in the PC tool?	The buffering storage interval of the communication unit is not used when connected to the PC tool and is not affected.	—
3	What is the relationship between the measurement cycle setting of the amplifier and the data acquisition interval of the PC tool?	The cycle in which the amplifier determines the measured value is different from the cycle in which the PC tool acquires data from the communication unit. For details, please refer to the page on the right.	
4	What kind of operation should I perform to reflect the setting parameters set in the amplifier itself in the tool?	Please reload the amplifier's settings by either of the following operations. (1)Execute the  button on the detailed setting screen. (Only the target CH is reflected.) (2) Click [Option] in the menu, and then click [Import Configurations from the Devices].	
5	When I set a threshold of 1000 or more for a short distance type from this tool, it shows 999.99 on the amplifier OLED screen, is this correct behavior?	In the case of the short distance type, when a threshold value of 1000 or higher is set using this tool, the amplifier OLED screen displays 999.99. This is due to the limitation that threshold values of ± 1000 or higher cannot be displayed on the OLED screen. There is no problem with the set value.	—

End of File